Agenthesis

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A read-only line-integrity oracle: it benchmarks a betting operator's prices against TxLINE's vig-free consensus, warns the instant a line is stale enough to get picked off, and settles every warning on-chain. You keep the book; it never touches it. Built on the TxLINE World Cup data layer by Onenept Studios.

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1. Abstract

In-play betting markets move fast, and the operators who post prices lose money at one specific moment: when their line is stale. The consensus re-prices on new information (a goal, a red card, a surge of danger) and a book that has not caught up is lifted at the old number. Agenthesis is a read-only agent that watches that gap. It benchmarks a watched price against TxLINE's de-margined (no-vig) consensus, classifies every reference-line move as a clean move to FOLLOW or an overreaction to FADE, warns before the pickoff, and grades every call against on-chain ground truth. It is not a bookmaker, a market-maker, or a managed-trading service; it is the neutral, provable benchmark that sits beside the book.

2. The problem: the stale line gets picked off

Adverse selection is the structural cost of quoting a price. Sharp money (bots, syndicates, faster books) exists to lift a mispriced line, and in-play is where the mispricing lives: the seconds around a goal, when the fair price has jumped and a lagging in-play number has not. The books most respected by sharps win on speed of price discovery; everyone slower leaks margin to stale-line abuse. Prediction-market makers face the identical phenomenon under the name loss-versus-rebalancing, where static automated prices become stale as information arrives and are picked off by better-informed flow. One phenomenon, two buyers. The question is the same: is my price stale right now, and in which direction am I exposed?

3. The idea: an independent, read-only benchmark

Agenthesis answers that question and stops. It emits a signal (a recommendation with a confidence and a pickoff-risk) and the operator's own rule-set decides whether to widen a margin, cut a limit, or suspend a market. We compute the decision; the book takes the action. That boundary is the entire product: it is why an unknown vendor's agent is something a compliance team will actually deploy, and why the tool carries no wagering or securities surface. The reference is not our opinion; it is TxLINE's de-margined consensus, so the benchmark is neutral by construction. Nothing about the operator's pricing model is required, replaced, or exposed.

4. The data layer: TxLINE

Everything Agenthesis does rests on one thing only TxLINE provides: a de-vig (de-margined) odds stream. The bookmaker margin is stripped out, so each side's price is already a clean implied probability: pRef = 1 / (price/1000). That is the whole trick - with the vig removed, a line move is no longer noise, it is a measurable shift in the TRUE price, which is what lets us separate a real move to follow from an overreaction to fade. No ordinary odds feed exposes this; it is why the product can only run on TxLINE. Two goals-settled families stream demargined today (Asian-handicap goals and over/under goals) and both resolve from the two on-chain goal counts, so every signal is settleable and verifiable.

TxLINE serves a second stream we depend on just as much: a granular possession tape alongside the scores feed (danger and high-danger possession, goal-imminent flags) that reads the attacking pressure seconds before the line jumps. Our goal_imminent signal is built entirely on that tape, and it is where the next generation of signals comes from - the more of the possession stream we read, the more we can flag before a price ever moves. So Agenthesis consumes two TxLINE streams, and only TxLINE streams. The feed is anchored on Solana and access is minted by a real on-chain subscribe transaction, so the reference's provenance is publicly verifiable.

Coverage is the natural next step. Today the signals are scoped to the goals markets that stream demargined; the more of the de-vig book TxLINE streams beyond goals (cards, corners, match-result, shots), the more of an operator's book Agenthesis can watch. Broader demargined coverage is a direct multiplier on how many of an operator's lines we can protect.

TxLINE endpoints used (server-held token: guest JWT + an on-chain Solana subscribe transaction -> apiToken, sent as Authorization: Bearer and X-Api-Token):

- GET /api/odds/stream - live de-margined (no-vig) odds, SSE; the core reference input.
- GET /api/scores/stream - live scores, match events, and the momentum tape, SSE.
- GET /api/scores/snapshot/{fixtureId} - final goals for outcome settlement.
- GET /api/scores/stat-validation - validateStat Merkle proof vs the on-chain daily-scores root.
- GET /api/fixtures/snapshot - live fixtures, team names, kickoff times.

5. The signal engine

The engine ingests odds and score frames and classifies each reference-line move, grounded in the market-microstructure literature:

- steam -> follow. THE PRIMARY EDGE. The market prices real news efficiently and momentum persists (Croxson & Reade; Moskowitz) - on our captures a flagged move held ~89% of the time. A clean move is true; a book that follows late is exactly the stale price a sharp lifts. Tighten toward the reference.
- overreaction -> hold / fade. The exception, not the rule. Bettors underreact to most goals and overreact only to surprising ones (Choi & Hui; De Bondt-Thaler), so only a minority overshoot and revert (~18% in our data). Default is hold; escalate to fade only on the surprise path, never on magnitude alone - big goal-moves are usually decisive and stick.
- goal_imminent -> suspend. A first-class signal off the momentum tape (high-danger possession / an explicit goal-imminent flag) that fires seconds before a goal lands, carrying a quantified goalProb = the calibrated P(goal <=120s); high-danger possession runs a measured 1.9x the base arrival rate. Our drift test found no tradeable pre-goal line move (the consensus already prices the danger), so the action is suspend/ widen only - the earliest notice a price is about to go stale.

Overreaction firing is sharpened by surprise: how far the goal moved the scoreline probability from its pre-event value. Steam and overreaction signals are scoped to the two on-chain-settleable goals markets, so nothing is emitted that cannot later be proven.

6. Grading: Fair Close Value and on-chain self-scoring

A call is right when the line behaves as the signal said it would - and for a FOLLOW that is not 'CLV is positive'. A follow is taken at fair value, so its expected closing-line value is ~0 by construction; grading it on CLV>0 would fail about half the continuations that were in fact correct. So the skill leg is Fair Close Value (FCV): the demargined fair probability at the +180s close. A follow or hold is right when the line HELD in the region it moved to (FCV stayed within +/-10pp of entry), because a book still quoting the old number is then left behind. A FADE keeps the reversion test: the overshoot came back. FCV resolves from odds alone, so it settles fast and with low variance; CLV is retained as an auxiliary diagnostic.

goal_imminent has no line to close against, so it is graded on a third axis entirely: goal ARRIVAL. We settle it against whether a goal actually landed inside the window and report the lift over the base arrival rate (~1.9x on our captures) - the honest proof for an anticipation signal is not 'the line moved' but 'the goal came disproportionately often'.

The outcome leg settles against the final goals on the TxLINE daily-scores Merkle root via a validateStat proof. The result is a public calibration ledger where the agent grades itself on-chain: follow/hold held-rate, fade reversion-rate, and goal_imminent arrival-lift per signal type and action, with per-match breadth and single-match concentration surfaced so a headline cannot hide behind one lucky match.

7. The read-only boundary

Agenthesis places no bet, moves no price, and holds no funds. The action is always the operator's. The Control Room makes the boundary visible: each signal, the gap between a watched book and the reference (the pickoff surface), and the action the operator's policy chose (widen, cut, or suspend). The policy is a rule-set the operator controls; we report which rule fired. This is also the answer to 'what if the agent is wrong?': it is wrong a knowable fraction of the time, and the design makes wrong cheap. Recommendations are confidence-weighted, the default under uncertainty is the safe action, and the operator sets the exposure envelope. It is a positive-expectation risk policy, not a must-be-right prediction.

8. Why it's adoptable: the independent referee

Incumbents already sell repricing: managed trading services and dynamic-pricing engines that adjust an operator's odds in real time. Agenthesis deliberately does not compete there. That lane is both the most contested and the one an operator is least willing to hand a startup, because it means giving up control of the book. The incumbents' structural weakness is that they are player and referee at once: they price your book, they may share your P&L, and they sell you the integrity feed, an unauditable black box. Agenthesis is the neutral referee they cannot be: no managed trading, no shared P&L, no conflict; read-only; and uniquely provable, because the track record settles on-chain. Verify-before-trust is the antidote to the black-box problem, and it is the one thing a non-anchored feed cannot offer.

9. Proof and verifiability

Every signal carries a proofHash tying it to the exact TxLINE frame it was derived from, reconcilable against a downloadable frame ledger (join on fixture and frame timestamp to confirm our reference matches yours). The Solana touchpoint is proof of access: a real on-chain subscribe transaction, signed with a wallet, mints the right to the TxLINE stream; that signature is a public, verifiable hash anyone can open on Solana Explorer. Outcome settlement anchors to the same chain via the daily-scores Merkle root.

10. The Operator API and SDK

The product is the API. Agenthesis is delivered as GET /api/v1/signals (an authenticated, versioned HTTP feed of read-only signals), alongside GET /api/v1/calibration (the provable track record) and GET /api/v1/control-room (the read-only boundary timeline). Every signal carries a proofHash, and a webhook pushes the identical payload from a persistent worker. An operator integrates in an afternoon, with no code to embed and nothing of their pricing model exposed. The SDK is an optional thin wrapper around the identical pure functions (the detector, the classifier, and the grader) for latency-sensitive consumers that run the classifier in-process, where a network round-trip cannot sit inside a millisecond pickoff loop. It is the exact code the API serves (SDK<->API parity): pure, deterministic, unit-tested, safe to place next to a live book.

11. Infrastructure: why this needs TxOdds

A production line-integrity signal is a latency game. The warning is only worth money if it beats the pickoff by milliseconds, which requires direct, co-located access to the TxLINE feed and low-latency infrastructure that only TxOdds can provision. The deterministic poll and replay in this build prove the logic on real captured frames; a live deployment is a different class of system. A win here is therefore the start of a continuing partnership (direct-feed and infrastructure support), not a finished artifact. The value compounds with every logged match, and the moat (an on-chain-provable calibration record) is one no non-anchored competitor can reproduce.

And the partnership runs both ways. Agenthesis is also a reason to be on TxLINE: any bookmaker or prediction market already taking the feed can bolt it on and instantly harden its line integrity (no new pricing model, no giving up the book), so it makes the de-vig feed worth more to the operators who buy it - an upgrade sitting on top of the data layer. Concretely, continued support means two things: low-latency direct access to both streams (the de-vig odds and the possession tape), and more of the demargined book beyond goals, so the shield can cover every line an operator quotes, not only the goals markets.

12. Responsible use

Agenthesis is a read-only research and risk-analytics layer built on de-margined data. It places no wagers, holds no funds, and moves no prices; the operator's rule-set takes every action. Fair Close Value and reversion are measures of pricing skill, not a promise of profit, and calibration over a replay does not guarantee live results. Nothing here is financial advice.